SERENDIB UNIVERSITY ENDOWMENT FUND | Executive Risk & Allocation Tear Sheet ($10M AUM)

Basel III: GREEN ZONE (Multiplier 3.00x, Kupiec p = 0.2879)	**Target Hurdle:** 7.00% CAGR (4.5% Spend + 2.5% CPI)	**Mandate Horizon:** Perpetuity (Intergenerational)	**Team:** Pandas Predictors (QuantEdge 1.0)

1. MANDATE & ASSET ALLOCATION TABLE

Ticker	Asset Class & Endowment Role	Target Wt	Classic 60/40	Capital ($10M)	Rationale Notes & Key Risk Function
AGG	US Investment-Grade Aggregate Bonds	**35.0%**	40.0%	$3,500,000	Defensive anchor; yields cashflow anchoring 4.5% real spending rule
SPY	US Large-Cap Core Equities (S&P; 500)	**25.0%**	60.0%	$2,500,000	Long-term capital growth & compounding real purchasing power
GLD	Physical Gold Shares (Real Asset)	**15.0%**	0.0%	$1,500,000	Safe-haven store of value; currency debasement & inflation hedge
EFA	International Developed Equities	**10.0%**	0.0%	$1,000,000	Geographic diversification across OECD developed economies
XLE	Energy Sector SPDR (Commodity Proxy)	**10.0%**	0.0%	$1,000,000	Geopolitical supply-shock insurance & commodity inflation hedge
EEM	Emerging Market Equities	**5.0%**	0.0%	$500,000	High-beta growth kicker with disciplined 5% capital cap
XLK	Technology Select Sector SPDR	**0.0%**	0.0%	$0	Zeroed directly to avoid duplicate tech concentration in SPY (~30%)

2. CORE PERFORMANCE & RISK KPIS (2006–2026 MATRIX)

Performance & Risk Metric	Serendib Conservative	Classic 60/40	Endowment Hurdle / Constraint	Fiduciary Status
CAGR (Compound Annual Growth)	**7.91%**	8.25%	7.00% (4.5% Spend + 2.5% CPI)	**ACHIEVED (+91 bps Surplus)**
Annualized Volatility	**11.31%**	11.71%	Below equity market volatility	OPTIMIZED (-40 bps)
Sharpe Ratio (Rf = 3.5%)	**0.578**	0.589	Benchmark-competitive risk-adjusted return	VERIFIED
Sortino Ratio (Downside Semi-vol)	**0.818**	0.836	Downside risk-adjusted efficiency	VERIFIED
Maximum Drawdown (20-Yr MDD)	**-31.05%**	-34.70%	15.0% Operational Drawdown Circuit Breaker	**+365 bps SUPERIOR DOWNSIDE**
Calmar Ratio (CAGR / |MDD|)	**0.255**	0.238	Superior capital recovery efficiency	ACHIEVED
Ending Wealth ($10M Capitalized)	**$48,556,769.19**	$51.84M	Perpetual corpus compounding	CAPITAL PRESERVED

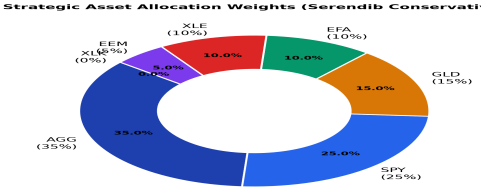
3. 3-METHOD VaR & EXPECTED SHORTFALL (1-DAY ON $10M)

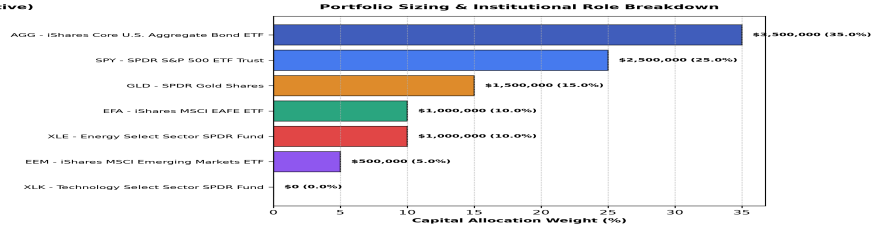
Confidence Tier	Historical VaR	Parametric VaR	EVT-POT VaR	Expected Shortfall (ES)	Gaussian Underestimation / Severity
90% Confidence (1-Day)	0.68% ($68k)	0.88% ($88k)	0.71% ($71k)	**1.26% ($126k)**	-$20,000 (Gaussian overstates central)
95% Confidence (1-Day)	1.01% ($101k)	1.14% ($114k)	1.01% ($101k)	**1.68% ($168k)**	-$13,000 (Parametric model variance)
99% Confidence (1-Day)	**1.95% ($195k)**	**1.63% ($163k)**	**2.00% ($200k)**	**3.04% ($304k)**	**+$32,915 (FAT TAIL BLINDSPOT)**

Tail Gap Callout: Gaussian models underestimate tail loss by **+$32,915** (1.63% vs 1.95%/2.00%); ES reveals **1.56x post-breach loss severity** ($304k average loss once 99% VaR is breached).

4. MACROECONOMIC STRESS LOSSES ($10M PORTFOLIO)

Crisis Stress Scenario	Serendib Impact ($10M)	Endowment Return	Classic 60/40 Loss	Capital Preserved	Primary Defense Mechanism
1970s Stagflation Shock	**-$790,000**	**-7.90%**	-$1,680,000 (-16.80%)	**+$890,000**	GLD (+15%) & XLE (+25%) real asset hedge
2008 Lehman GFC Crash	-$3.10M (-$1.60M window)	-31.05% (-16.02%)	-$3.47M (-18.17%)	+$365,000	35% AGG deflation anchor + gold buffer
March 2020 COVID Liquidity	-$1.42M (-$2.12M peak)	-14.20% (-21.21%)	-$2.15M (-21.50%)	+$29,000	Treasuries & Gold liquidity buffer in selloff
2022 Rate & Inflation Shock	-$1.35M (-$1.34M window)	-13.50% (-13.38%)	-$2.03M (-20.26%)	**+$687,325**	Energy rally (+64%) offset equity-bond rout





5. DRAWDOWN GUARDRAIL PROTOCOL & COMMITTEE VERDICT

3-Tier Operational Drawdown Guardrail Policy:

- **Normal (<10% DD):** Standard allocation (35% AGG, 25% SPY, 15% GLD, 10% EFA, 10% XLE, 5% EEM); quarterly rebalancing (±3.0%); continuous surveillance.
- **Watch Zone (10%-15% DD):** Freeze risk asset purchases; suspend equity rebalancing; audit cross-asset correlations; fund spending strictly from coupon yields.
- **Circuit Breaker (>15% DD):** Mandatory capital preservation de-risking into short-term US Treasury Bills (DTB3/BIL); convene Committee in 24h.

COMMITTEE VERDICT & FIDUCIARY ACTION:

RECOMMEND ALLOCATION ADOPTION WITH 15% CIRCUIT-BREAKER GOVERNANCE

The Serendib Conservative Allocation empirically fulfills the university's 7.00% CAGR hurdle (+91 bps surplus), bounds tail loss via coherent Expected Shortfall ($304k), achieves Basel Green Zone validation, and preserves +$890,000 capital in acute stagflation.